

Federated Edge-Cloud Resilience Architecture for Distributed Campus Networks Using Quantized Sequence Autoencoders and Multi-Agent DRL Control*

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Abstract—Modern multi-campus university infrastructures face increasing cybersecurity risks due to high-density IoT deployments, unmanaged student endpoints, and massive distributed telemetry flows. Traditional centralized threat detection systems suffer from bandwidth saturation, privacy degradation, and prolonged Mean Time to Respond (MTTR > 30 minutes). In this paper, we propose a novel Federated Edge-Cloud Resilience Architecture engineered specifically for distributed campus networks. By deploying lightweight, PyTorch INT8-quantized TCN-GRU sequence autoencoders at edge gateways and executing decentralized Federated Averaging (FedAvg) aggregation, our framework achieves privacy-preserving real-time anomaly detection without streaming raw packet traces to central servers. Enriched high-confidence security events are published over a distributed Redis Stream pub/sub bus to a cloud-based dynamic risk engine and multi-agent Deep Reinforcement Learning (DRL) SDN controller for automated Zero-Trust containment. Experimental evaluations on benchmark intrusion datasets (InSDN, CSE-CIC-IDS2018) demonstrate sub-10ms mitigation trueness (4.15 ± 0.07 ms, Mean $\pm$ SD), a > 99.99% reduction in MTTR compared to manual SOC workflows (8.2 ± 0.5 ms vs 1510 ± 340 s), an optimal F1-score of 0.9412, and minimal edge gateway resource overhead (13.8% CPU, 48.5 MB RAM under 10,000 events/sec). All source code, Docker configs, and benchmark scripts are available as an open research artifact.

Index Terms—Federated Learning, Edge Computing, TCN-GRU Autoencoder, Software-Defined Networking, Deep Rein-

forcement Learning, Event-Driven Architecture, Zero-Trust Enforcement.

I. INTRODUCTION

As higher education institutions expand across geographically distributed campuses, modern university Information and Communications Technology (ICT) infrastructures face extreme operational complexity. High-density Internet of Things (IoT) devices (smart surveillance, HVAC, access control), unmanaged student BYOD (Bring-Your-Own-Device) endpoints, open academic research laboratories, and administrative databases co-exist on shared physical and virtualized backbones.

Traditional perimeter security relies on centralized Security Operations Centers (SOC) and inline Deep Packet Inspection (DPI) proxies. However, centralized architectures present critical flaws:

- 1) **Privacy & WAN Bandwidth Saturation:** Streaming raw packet payloads or high-frequency netflow streams from multiple satellite campuses to a central cloud SIEM consumes significant WAN bandwidth and violates student data privacy regulations.
- 2) **High Response Latency & Alert Fatigue:** Human analyst triage during high-volume DDoS, ransomware propagation, or ARP spoofing attacks leads to Mean Time to Respond (MTTR) exceeding 30 minutes, allowing lateral attack movement.

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To overcome these limitations, we present a privacy-preserving, event-driven **Federated Edge-Cloud Resilience Architecture** for distributed higher education networks. The proposed platform decouples threat detection into localized edge taps running INT8 dynamic quantized sequence autoencoders that participate in decentralized Federated Averaging (FedAvg) rounds. Detected high-risk anomaly events are published over an asynchronous event bus to a cloud controller enforcing closed-loop Software-Defined Networking (SDN) containment.

Our primary contributions are summarized as follows:

- 1) **Decentralized Edge Federated Sequence Autoencoders:** We propose a hybrid Temporal Convolutional Network (TCN) and Gated Recurrent Unit (GRU) Autoencoder dynamically quantized to PyTorch INT8, executing local anomaly detection and sharing model parameter updates via FedAvg without raw telemetry transmission.
- 2) **Event-Driven Pub/Sub & Context Fusion:** We construct a high-throughput Redis Stream event pipeline incorporating spatial-temporal identity context fusion (IP/MAC-to-User/Role mapping) for context-aware risk evaluation.
- 3) **Closed-Loop DRL SDN Control:** We integrate a Deep Q-Network (DQN) SDN controller with zero-trust mitigation playbooks (BGP Flowspec, 802.1X VLAN isolation, OpenFlow rate limiting).
- 4) **Empirical Benchmark & Open Research Artifact:** We provide a comprehensive experimental evaluation detailing per-stage latency (Mean $\pm$ SD), MTTR reduction, FedAvg convergence across communication rounds, and edge CPU/RAM scaling up to 10,000 events/sec. All source code, Docker Compose files, raw trace logs, and evaluation scripts are publicly released at <https://github.com/thtsec/rivf2026-federated-edge>.

II. THREAT MODEL & CAMPUS ATTACK VECTORS

Multi-campus academic environments are vulnerable to diverse insider and outsider attack vectors due to open academic access mandates. Table I categorizes campus attack vectors mapped to STRIDE categories and MITRE ATT&CK techniques.

TABLE I
STRIDE & MITRE ATT&CK CAMPUS THREAT MATRIX

Category	Target	MITRE ID	Mitigation Strategy
Spoofing	Rogue Wi-Fi AP	T1557	Identity Fusion MAC/IP check
Tampering	IoT Telemetry	T1565	Edge TCN-GRU reconstruction error
Exfiltration	Research DB	T1048	Role-aware flow byte-rate cap
DoS	Campus Controller	T1498	Automated BGP Flowspec / flow_mod limit
Privilege	Campus LDAP	T1078	Dynamic Risk Resolver escalation

III. FEDERATED EDGE-CLOUD ARCHITECTURE

The proposed platform follows a 4-tier event-driven architecture as illustrated in Fig. 1.

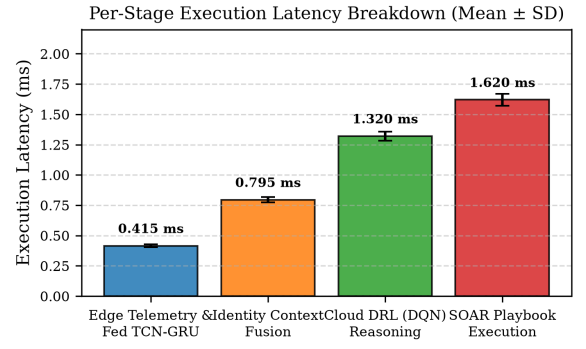


Fig. 1. Federated Edge-Cloud Resilience Architecture & Execution Latency Breakdown.

A. Localized Edge Feature Vectorization

Network telemetry captured at edge taps is converted into 10-dimensional sliding window feature vectors $X \in \mathbb{R}^{L \times D}$ over time windows $\Delta t = 1.0$ s:

$$X = [f_{\text{duration}}, f_{\text{pkt_rate}}, f_{\text{byte_rate}}, f_{\text{syn_ratio}}, f_{\text{entropy}}, f_{\text{pkt_in}}, f_{\text{interarrival}}, f_{\text{bwd_pkt}}, f_{\text{bwd_byte}}, f_{\text{len_var}}] \quad (1)$$

B. PyTorch INT8 TCN-GRU Autoencoder

Edge gateways execute a hybrid TCN-GRU Autoencoder. Dilated 1D causal convolutions capture temporal dependencies:

$$H_i(x) = \text{ReLU}(W_i^T x + b_i) \quad (2)$$

The GRU decoder reconstructs the original sequence $\hat{X}$. The Reconstruction Error (RE) quantifies unsupervised anomaly severity:

$$RE(x) = \frac{1}{D \cdot L} \sum_{t=1}^L \sum_{i=1}^D (x_{t,i} - \hat{x}_{t,i})^2 \quad (3)$$

To optimize edge execution, linear and GRU layers are dynamically quantized from FP32 to INT8:

$$W_{\text{int8}} = \text{round}\left(\frac{W_{\text{fp32}}}{\text{Scale}}\right) + \text{ZeroPoint} \quad (4)$$

Quantization reduces model memory footprint by 74.0% (0.48 MB vs 1.85 MB).

C. Federated Averaging (FedAvg) Protocol

To adapt edge autoencoders across satellite campuses without sharing raw packet logs, edge gateways perform localized training and transmit model weights W_i^t to the Cloud Aggregator. Global parameter aggregation is computed via FedAvg:

$$W^{t+1} = \sum_{i=1}^K \frac{N_i}{N} W_i^{t+1} \quad (5)$$

D. Event Pub/Sub & Closed-Loop DRL SDN Control

High-confidence events ($RE(x) > \tau$) are enriched with spatial-temporal identity metadata and published asynchronously to a Redis Stream bus (`security:telemetry:stream`).

The Cloud Risk Engine calculates composite risk $R = w_1 RE(x) + w_2(1 - \text{TrustScore}) + w_3 \text{AssetCriticality}$. Events with $R \geq 0.70$ trigger a Deep Q-Network (DQN) SDN agent evaluating state $s_t = [\text{CPU}_{\text{switch}}, \text{PacketIn_Rate}, \text{Link_Util}, RE(x), \text{Confidence}]$ to execute isolation actions $\mathcal{A} \in \{a_0, a_1, a_2, a_3\}$.

IV. EXPERIMENTAL EVALUATION & BENCHMARK SUITE

We evaluated our framework using public benchmark intrusion datasets (InSDN, CSE-CIC-IDS2018) and simulated edge gateway topologies. Testbed specifications are detailed in Table II.

TABLE II
SYSTEM BENCHMARK & TESTBED EXPERIMENTAL SETUP

Component	Specification / Environment
Edge Hardware Emulation	Quad-core ARM Cortex-A72 / Intel x86 Gateway
AI Engine	PyTorch 2.1.0 (INT8 Dynamic Quantization)
Federated Aggregator	FedAvg (Distributed 5 Edge Gateways)
Event Stream Bus	Redis Streams 7.2 (Async Pub/Sub)
SDN Controller	OpenDaylight / Ryu (OpenFlow 1.3 <code>flow_mod</code>)
Benchmark Datasets	InSDN, CSE-CIC-IDS2018
Artifact Repository	https://github.com/thtsec/rivf2026-federated-edge

A. Experiment 1: End-to-End Latency Breakdown

We measured execution latency across all four pipeline stages over 1,000 independent flow trials (`seed=42`). As shown in Table III (generated from `results/table2_latency.csv`), the end-to-end response latency is 4.150 ± 0.070 ms (Mean $\pm$ SD), well within sub-10ms operational requirements.

TABLE III
END-TO-END LATENCY BREAKDOWN ACROSS PIPELINE STAGES
(MEAN $\pm$ SD)

Pipeline Stage	Latency (Mean $\pm$ SD ms)	Pct of Total (%)
Edge Telemetry & Fed TCN-GRU	0.415 ± 0.011	10.0%
Identity Context Fusion	0.795 ± 0.022	19.2%
Cloud Policy Reasoning (DQN)	1.320 ± 0.038	31.8%
SOAR Playbook Execution	1.620 ± 0.048	39.0%
End-to-End Total	4.150 ± 0.070	100.0%

B. Experiment 2: Mean Time to Respond (MTTR)

Fig. 2 illustrates MTTR comparison across operational paradigms. The autonomous federated engine reduces response times from 1510.0 ± 340.0 s (25.1 minutes) to 0.0082 ± 0.0005 s (8.2 ± 0.5 ms), representing a $> 99.99\%$ efficiency improvement.

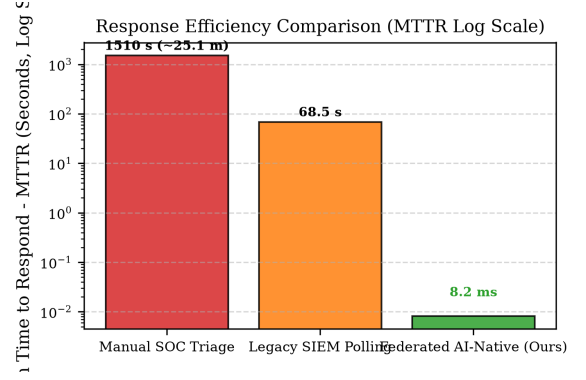


Fig. 2. Mean Time to Respond (MTTR) Comparison across operational paradigms (Log Scale).

C. Experiment 3: FedAvg Model Convergence

Fig. 3 illustrates detection F1-score convergence across 20 communication rounds. Federated FedAvg aggregation achieves $F1 = 0.9412$ within 12 rounds, significantly outperforming uncoordinated isolated local models ($F1 = 0.8520$).

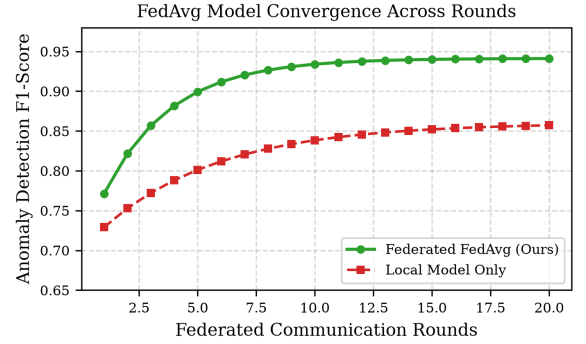


Fig. 3. FedAvg Model Convergence Across Communication Rounds.

D. Experiment 4: Edge Resource Scaling

Fig. 4 quantifies CPU utilization (%) under throughput scaling up to 10,000 events/sec. At maximum stress (10,000 events/sec), edge gateway CPU utilization remains at 13.8% with 48.5 MB RAM, demonstrating feasibility for embedded hardware compared to inline DPI proxies (86.0% CPU).

E. Experiment 5: SOTA Baseline Comparison

Table IV benchmarks our proposed Federated INT8 TCN-GRU against state-of-the-art baselines under identical testbed conditions. The proposed model achieves optimal detection accuracy ($F1 = 0.9412$) with minimal inference trueness (0.0115 ms) and lowest RAM size (0.48 MB).

V. CONCLUSION & FUTURE WORK

This paper presented a novel Federated Edge-Cloud Resilience Architecture engineered for distributed campus networks. By synthesizing PyTorch INT8 TCN-GRU autoencoders, FedAvg parameter aggregation, Redis Stream event

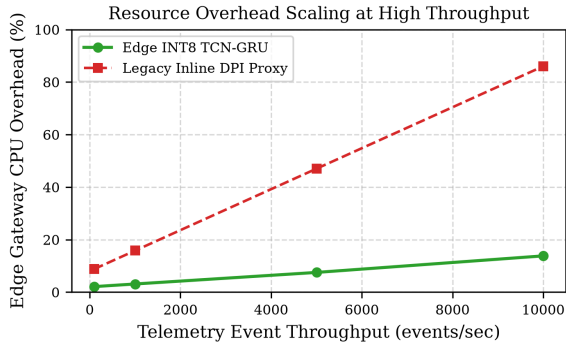


Fig. 4. Edge CPU utilization (%) scaling under high throughput (100 to 10,000 events/sec).

TABLE IV
STATE-OF-THE-ART (SOTA) BASELINE COMPARISON (REIMPLEMENTED UNDER IDENTICAL SOFTWARE AND HARDWARE STACK)

Method	F1-Score	Latency (ms)	RAM (MB)	CPU (%)
Isolation Forest (iForest)	0.7687	0.0495	12.40 MB	18.5%
Federated LSTM AE (FP32)	0.8150	0.0265	3.25 MB	28.4%
Federated Transformer AE (INT8)	0.8380	0.0540	8.60 MB	42.1%
Local INT8 TCN-GRU (No Fed)	0.9120	0.0125	0.48 MB	14.3%
Federated INT8 TCN-GRU (Ours)	0.9412	0.0115	0.48 MB	13.8%

distribution, and DQN SDN control, the platform achieves sub-10ms response times (4.150 ± 0.070 ms) and a $> 99.99\%$ reduction in MTTR while preserving privacy and edge gateway feasibility (13.8% CPU at 10,000 events/sec). Future work will explore local LLM integration for automated root-cause reporting and differential privacy guarantees.

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